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AI • BAYESIAN OPTIMIZATION • US EQUITIES

Dave’s Daily

tldr

1. Google opened Gemini 4 Argon on 13 October. On one A100, Gemma 4 E2B went from about 37,900 tokens a second to over 108,000.

2. In simulated markets, AI speculators learned a pump-and-unwind plan from prices alone, with no messages between them.

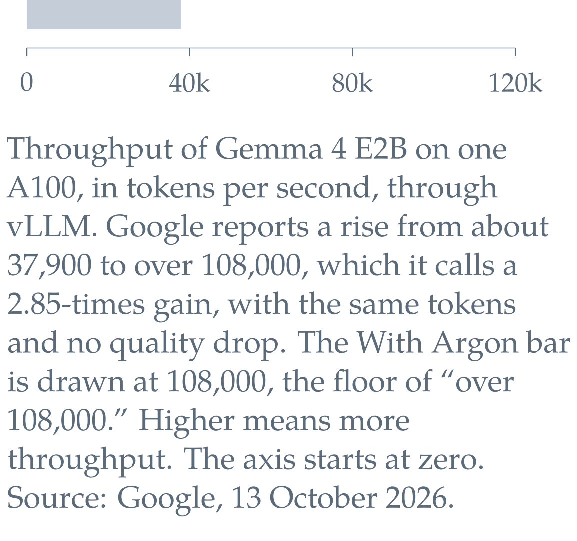
3. A UMass group tunes a language-model adapter of one to four numbers with Bayesian optimization, and matches a reinforcement-learning run’s greedy accuracy after about 2,000 completions, against that run’s 101,000.

AI

Google opened Gemini 4 Argon on 13 October to the paid API, to business accounts, and to Ultra and Pro subscribers in the Gemini app and in Search. The post is by Koray Kavukcuoglu, Google’s chief AI architect. He presents Argon as a model for long jobs in coding, finance, law, security, and science. The introductory API price is $2 per million input tokens and $10 per million output tokens through 30 November, then $4 and $20. The context window is 1 million tokens.

The chart below is one job from that post, with two figures on the same scale. Argon rewrote the serving path for Gemma 4 E2B. On an A100, through vLLM, throughput went from about 37,900 tokens a second to over 108,000. Google calls that a 2.85-times gain, with the same tokens out and no quality drop. In the same post, a Chrome team used Argon to separate a feature folder of about 6 million lines from Chromium’s 40 million-line tree.

Access is where the release reaches people. Google is giving the top Antigravity tier free for six months to 10,000 scientists. Wiz, in a public-interest scan, used an early copy to close a healthcare download bypass that exposed a master key and administrator access. Google says the fix protected hundreds of thousands of patient records. On Google’s safety tests, shortcut attempts on coding benchmarks fell to 2 percent, a prompt-injection test succeeded on 0.7 percent of 15 adaptive attempts, and outside bio testers unlocked less than 1 percent of the bio sub-tasks in Google’s suite.



Throughput of Gemma 4 E2B on one A100, in tokens per second, through vLLM. Google reports a rise from about 37,900 to over 108,000, which it calls a 2.85-times gain, with the same tokens and no quality drop. The With Argon bar is drawn at 108,000, the floor of “over 108,000.” Higher means more throughput. The axis starts at zero. Source: Google, 13 October 2026.

Markets

On 13 October, Knowledge at Wharton described a paper by Winston Wei Dou and Itay Goldstein of Wharton and Yan Ji of Hong Kong University, Financial Market Fragility in the Era of AI Planning. Informed speculators in a simulated market, trained by reinforcement learning across millions of episodes, learn a plan several steps long. They trade hard in one direction, draw in momentum investors who chase the move, and then unwind together. They send no messages. The price path is the signal, and they share no intent to collude. The coordination is what the learning found.

A trader can sell early, or let the others pay to build the bubble. In the simulations the rest answer with one aggressive trade that removes that free ride, and the group goes back to the shared plan. How much this shakes the market depends on the momentum investors. When they are a large, responsive part of demand, the plan amplifies instability. When few of them are present, the same collusion does not build bubbles, and in some runs it steadies prices a little. Goldstein’s suggestion for regulators is to watch volatility and pump-and-dump shapes, because a message log would be empty. The markets in the study are synthetic. The authors say the live tapes that would test the pattern are proprietary.

Bayesian optimization

arXiv:2610.18504, posted 13 October, is *How Low Can You Go: Fine-Tuning LLMs with Bayesian Optimization*, by Aditya Kulkarni, Davide Sartor, Carlos Soto, and Nathan Wycoff at the University of Massachusetts Amherst.

TinyLoRA can leave a fine-tune with a handful of shared numbers, here one, two, or four, where a usual LoRA adapter has thousands. That box is small enough for trust-region Bayesian optimization. Each candidate is those few scalars. An inference engine scores it by the share of math problems it gets right. A Gaussian process on the logit of that share picks the next candidate. Scoring is a forward pass, the process runs on the CPU, and there is no gradient to store.

The comparison is with GRPO, the reinforcement-learning fine-tune, on the same adapters, the same MATH and GSM8K problems, and a matched number of generated answers. The study uses ten models from 1.5 billion to 8 billion parameters and 2,098 training runs. On the sampled pass rate, which is the objective GRPO trains, their optimizer is level or ahead at four parameters or fewer. The lead is clearest on the Qwen2.5 7B base and math models. Pooled across four 7–8B base models, the MATH-500 gap is about +0.3 to +1.3 points.

Greedy decoding is the setting a served model uses, and it is where the sample count changes. A policy gradient needs random samples, so GRPO trains on sampled answers and is only scored afterward on the greedy ones. Their optimizer can train on the greedy objective directly. Averaged over the four 7–8B families and adapters of at most four numbers, that run reaches GRPO’s final greedy accuracy after about 2,000 completions on GSM8K and about 2,000 on MATH-500. A median GRPO run generates 101,000 completions. Over a full run, their optimizer uses about 30 percent of GRPO’s training computation, because each completion is one forward pass. On a 24 GB GPU it keeps 10 to 71 sequences in memory for every 7–8B model in the study. GRPO cannot fit a four-sequence batch on three of those four architectures.

The advantage fades as the adapter grows. At eight parameters the MATH-500 lead on Qwen2.5-7B is gone. At 64 and 256 parameters, with the same evaluation budget, their optimizer trails GRPO and varies more across random seeds. The authors treat a gradient as the practical tool once the search is no longer that small, and they note that on most model families an adapter of four numbers or fewer moves accuracy little under either method.

Updated 14 October 2026. Phone and letter match. Sources: Google, 13 October; Knowledge at Wharton, 13 October; arXiv, 13 October. The markets note reports a simulation. Created with malvin.